

Striatal MSN Multicompartmental Model Parameters

This document catalogs the biophysical, morphological, and kinetic parameters of the reconstructed striatal Medium Spiny Neuron (MSN) model, based on the classic publication:

Wolf et al. (2005) - *Computational Model of the Striatal Medium Spiny Neuron: Biophysical Properties and State Transitions* (J. Neurosci 25:9080–9095).

1. Calcium Channel Biophysics & Kinetics

All voltage-gated calcium channels (VGCCs) use the **Goldman-Hodgkin-Katz (GHK)** current equation to calculate the calcium flux, which is driven by concentration gradients.

CaQ Channel (Q-type)

- **Permeability (Somatic):** $\bar{p} = 6.0 \times 10^{-6}$ cm/s
 - **State Variables:** m (activation, squared)
 - **Inactivation:** None modeled
 - **Current Formula:** $i_{Ca} = \bar{p} \cdot m^2 \cdot ghk(v, [Ca]_i, [Ca]_o)$
 - **Steady-State Gating:** $m_{\infty} = \frac{1}{1 + e^{-\frac{v + 9.0}{6.6}}}$
 - **Time Constant:** $\tau_m = 0.377$ ms (constant)
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CaN Channel (N-type)

- **Permeability (Somatic):** $\bar{p} = 1.0 \times 10^{-5}$ cm/s
- **State Variables:** m (activation, squared), h (inactivation)
- **Inactivation Fraction (a):** 0.21 (fraction of current that inactivates; $1 - a = 0.79$ is persistent)
- **Current Formula:** $i_{\text{Ca}} = \bar{p} \cdot m^2 \cdot (a \cdot h + 1 - a) \cdot \text{ghk}(v, [\text{Ca}]_i, [\text{Ca}]_o)$
- **Steady-State Gating:**

$$- m_{\infty} = \frac{1}{1 + e^{-\frac{V+8.7}{7.4}}}$$

$$- h_{\infty} = \frac{1}{1 + e^{-\frac{V+74.8}{6.5}}}$$

- **Gating Kinetics:**

$$- \alpha_m = \frac{0.1157 \cdot (V+17.19)}{e^{\frac{V+17.19}{15.22}} - 1}$$

$$- \beta_m = 1.15 \cdot e^{\frac{V}{23.82}}$$

$$- \tau_m = \frac{1}{\alpha_m + \beta_m}$$

$$- \tau_h = 23.33 \text{ ms (constant)}$$

CaR Channel (R-type)

- **Permeability (Somatic):** $\bar{p} = 2.6 \times 10^{-5}$ cm/s
- **State Variables:** m (activation, cubed), h (inactivation)
- **Current Formula:** $i_{\text{Ca}} = \bar{p} \cdot m^3 \cdot h \cdot \text{ghk}(v, [\text{Ca}]_i, [\text{Ca}]_o)$
- **Steady-State Gating:**

$$- m_{\infty} = \frac{1}{1 + e^{-\frac{V+10.3}{6.6}}}$$

$$- h_{\infty} = \frac{1}{1 + e^{-\frac{V+33.3}{17.0}}}$$

- **Gating Kinetics:**

$$- \tau_m = 1.7 \text{ ms (constant)}$$

$$- \tau_h: \text{Lookup Table (linear interpolation between } -30 \text{ mV and } 10 \text{ mV)}$$

CaR Inactivation Gating Time Constant (τ_h) Lookup Table

Membrane Potential V (mV)	τ_h (ms)
$V \leq -30.0$	33.33
-20.0	21.67
-10.0	11.67
0.0	10.00
$V \geq 10.0$	6.67

CaT Channel (T-type)

- **Permeability (Somatic):** $\bar{p} = 4.0 \times 10^{-7}$ cm/s
- **State Variables:** m (activation, cubed), h (inactivation)
- **Current Formula:** $i_{Ca} = \bar{p} \cdot m^3 \cdot h \cdot \text{ghk}(v, [\text{cal}]_i, [\text{cal}]_o)$
- **Steady-State Gating:**

$$- m_{\infty} = \frac{1}{1 + e^{-\frac{V + 51.73}{6.53}}}$$

$$- h_{\infty} = \frac{1}{1 + e^{-\frac{V + 80.0}{6.7}}}$$

- **Gating Kinetics:** τ_m and τ_h are determined by linear interpolation of a lookup table between -65 mV and 10 mV (step size 5 mV).

CaT Gating Time Constants (τ_m, τ_h) Lookup Table

CaL12 Channel (CaV1.2 L-type)

- **Permeability (Somatic):** $\bar{p} = 6.7 \times 10^{-6}$ cm/s
- **State Variables:** m (activation, squared), h (inactivation)
- **Inactivation Fraction (a):** 0.17
- **Current Formula:** $i_{Ca} = \bar{p} \cdot m^2 \cdot (a \cdot h + 1 - a) \cdot \text{ghk}(v, [\text{cal}]_i, [\text{cal}]_o)$
- **Steady-State Gating:**

Membrane Potential V (mV)	τ_m (ms)	τ_h (ms)
$V \leq -65.0$	6.73	127.33
-60.0	6.73	69.33
-55.0	4.37	54.00
-50.0	2.90	43.00
-45.0	2.27	39.67
-40.0	1.87	35.67
-35.0	1.47	35.67
-30.0	1.27	35.67
-25.0	1.20	36.00
-20.0	1.10	36.33
-15.0	1.20	36.33
-10.0	1.20	36.67
-5.0	1.10	36.67
0.0	1.10	36.67
5.0	1.10	36.67
$V \geq 10.0$	1.10	36.67

$$- m_{\infty} = \frac{1}{1+e^{-\frac{V+8.9}{6.7}}}$$

$$- h_{\infty} = \frac{1}{1+e^{-\frac{V+13.4}{11.9}}}$$

• **Gating Kinetics:**

$$- \alpha_m = \frac{0.1194 \cdot V}{e^{\frac{V}{9.005}} - 1}$$

$$- \beta_m = 2.97 \cdot e^{\frac{V}{31.4}}$$

$$- \tau_m = \frac{1}{\alpha_m + \beta_m}$$

$$- \tau_h = 14.77 \text{ ms (constant)}$$

CaL13 Channel (CaV1.3 L-type)

- **Permeability (Somatic):** $\bar{p} = 4.25 \times 10^{-7} \text{ cm/s}$
- **State Variables:** m (activation, squared), h (inactivation)
- **Inactivation Fraction (a):** 0.17

- **Current Formula:** $i_{Ca} = \bar{p} \cdot m^2 \cdot (a \cdot h + 1 - a) \cdot \text{ghk}(v, [\text{cal}]_i, [\text{cal}]_o)$

- **Steady-State Gating:**

$$- m_{\infty} = \frac{1}{1 + e^{-\frac{V+33.0}{6.7}}}$$

$$- h_{\infty} = \frac{1}{1 + e^{\frac{V+13.4}{11.9}}}$$

- **Gating Kinetics:**

$$- \alpha_m = \frac{0.1194 \cdot V}{e^{\frac{V}{9.005}} - 1}$$

$$- \beta_m = 2.97 \cdot e^{\frac{V}{31.4}}$$

$$- \tau_m = \frac{1}{\alpha_m + \beta_m}$$

$$- \tau_h = 14.77 \text{ ms (constant)}$$

2. Sodium & Potassium Channels

NaF Channel (Fast Transient Sodium)

- **Maximum Conductance (Somatic):** $\bar{g} = 1.5 \text{ S/cm}^2$
- **State Variables:** m (activation, cubed), h (inactivation)
- **Current Formula:** $i_{Na} = \bar{g} \cdot m^3 \cdot h \cdot (V - E_{Na})$
- **Steady-State Gating:**

$$- m_{\infty} = \frac{1}{1 + e^{-\frac{V+23.9}{11.8}}}$$

$$- h_{\infty} = \frac{1}{1 + e^{\frac{V+62.9}{10.7}}}$$

- **Gating Kinetics:** τ_m and τ_h are determined by linear interpolation of a lookup table between -80 mV and 30 mV (step size 10 mV).

NaF Gating Time Constants (τ_m, τ_h) Lookup Table

Membrane Potential V (mV)	τ_m (ms)	τ_h (ms)
$V \leq -80.0$	0.023	0.433
-70.0	0.030	0.433
-60.0	0.037	0.433
-50.0	0.043	0.433
-40.0	0.067	0.433
-30.0	0.107	0.433
-20.0	0.053	0.283
-10.0	0.050	0.167
0.0	0.040	0.150
10.0	0.027	0.107
20.0	0.020	0.100
$V \geq 30.0$	0.020	0.093

NaP Channel (Persistent Sodium)

- **Maximum Conductance (Somatic):** $\bar{g} = 4.0 \times 10^{-5} \text{ S/cm}^2$
- **State Variables:** m (activation), h (slow inactivation)
- **Current Formula:** $i_{\text{Na}} = \bar{g} \cdot m \cdot h \cdot (V - E_{\text{Na}})$
- **Steady-State Gating:**

$$- m_{\infty} = \frac{1}{1 + e^{-\frac{V+52.6}{4.6}}}$$

$$- h_{\infty} = \frac{1}{1 + e^{\frac{V+48.8}{10}}}$$

- **Gating Kinetics:**

$$- \tau_m:$$

$$\tau_m = \begin{cases} 0.025 + 0.14 \cdot e^{\frac{V+40.0}{10.0}} & \text{if } V < -40.0 \text{ mV} \\ 0.020 + 0.145 \cdot e^{-\frac{V+40.0}{10.0}} & \text{if } V \geq -40.0 \text{ mV} \end{cases}$$

$$- \tau_h: \text{Lookup Table (linear interpolation between } -100 \text{ mV and } 10 \text{ mV)}$$

NaP Inactivation Gating Time Constant (τ_h) Lookup Table

Membrane Potential V (mV)	τ_h (ms)
$V \leq -100.0$	1500.00
-90.0	1583.33
-80.0	1733.33
-70.0	2033.33
-60.0	2100.00
-50.0	1666.67
-40.0	1416.67
-30.0	1166.67
-20.0	1000.00
-10.0	900.00
0.0	833.33
$V \geq 10.0$	700.00

KAf Channel (Fast A-type Potassium)

- **Maximum Conductance (Somatic):** $\bar{g} = 0.225 \text{ S/cm}^2$
- **State Variables:** m (activation, squared), h (inactivation)
- **Current Formula:** $i_K = \bar{g} \cdot m^2 \cdot h \cdot (V - E_K)$
- **Steady-State Gating:**

$$- m_{\infty} = \frac{1}{1 + e^{-\frac{V + 10.0}{17.7}}}$$

$$- h_{\infty} = \frac{1}{1 + e^{\frac{V + 75.6}{10.0}}}$$

- **Gating Kinetics:**
 - $\tau_h = 4.67 \text{ ms}$ (constant)
 - τ_m : Lookup Table (linear interpolation between -40 mV and 50 mV)

KAf Activation Gating Time Constant (τ_m) Lookup Table

Membrane Potential V (mV)	τ_m (ms)
$V \leq -40.0$	0.600
-30.0	0.367
-20.0	0.333
-10.0	0.333
0.0	0.300
10.0	0.267
20.0	0.300
30.0	0.300
40.0	0.300
$V \geq 50.0$	0.267

KAs Channel (Slow A-type Potassium)

- **Maximum Conductance (Somatic):** $\bar{g} = 0.0104 \text{ S/cm}^2$
- **State Variables:** m (activation, squared), h (inactivation)
- **Inactivation Fraction (a):** 0.996 (fraction of current that inactivates; $1 - a = 0.004$ is persistent)
- **Current Formula:** $i_K = \bar{g} \cdot m^2 \cdot (a \cdot h + 1 - a) \cdot (V - E_K)$
- **Steady-State Gating:**

$$- m_\infty = \frac{1}{1 + e^{-\frac{V+27.0}{16.0}}}$$

$$- h_\infty = \frac{1}{1 + e^{\frac{V+33.5}{21.5}}}$$

- **Gating Kinetics:**

$$- \tau_m = 0.378 + 9.91 \cdot e^{-\left(\frac{V+34.3}{30.1}\right)^2}$$

$$- \alpha_h = e^{-\frac{V+90.96}{29.01}}$$

$$- \beta_h = e^{\frac{V+90.96}{100.0}}$$

$$- \tau_h = \frac{1097.4}{\alpha_h + \beta_h}$$

Kir Channel (Inwardly Rectifying Potassium)

- **Maximum Conductance (Somatic):** $\bar{g} = 1.4 \times 10^{-4} \text{ S/cm}^2$
- **State Variables:** m (activation)
- **Current Formula:** $i_K = \bar{g} \cdot m \cdot (V - E_K)$
- **Steady-State Gating:** $m_\infty = \frac{1}{1 + e^{\frac{V + 82.0}{-13.0}}}$
- **Gating Kinetics:** τ_m is determined by linear interpolation of a lookup table between -100 mV and 0 mV .

Kir Activation Gating Time Constant (τ_m) Lookup Table

Membrane Potential V (mV)	τ_m (ms)
$V \leq -100.0$	7.465
-90.0	8.000
-80.0	9.435
-70.0	10.755
-60.0	12.120
-50.0	13.795
-40.0	15.385
-30.0	14.285
-20.0	11.765
-10.0	8.890
$V \geq 0.0$	8.000

Krp Channel (Slowly Inactivating Delayed Rectifier Potassium)

- **Maximum Conductance (Somatic):** $\bar{g} = 0.001 \text{ S/cm}^2$
- **State Variables:** m (activation), h (inactivation)
- **Inactivation Fraction (a):** 0.70
- **Current Formula:** $i_K = \bar{g} \cdot m \cdot (a \cdot h + 1 - a) \cdot (V - E_K)$
- **Steady-State Gating:**

$$- m_{\infty} = \frac{1}{1+e^{-\frac{V+13.5}{11.8}}}$$

$$- h_{\infty} = \frac{1}{1+e^{-\frac{V+54.7}{18.6}}}$$

- **Gating Kinetics:** τ_m and τ_h are determined by linear interpolation of a lookup table between -100 mV and 50 mV (step size 5 mV).

Krp Gating Time Constants (τ_m, τ_h) Lookup Table

[h]

3. Calcium-Activated Potassium Channels

In medium spiny neurons, calcium-activated potassium channels are selectively activated by calcium influx from specific high-voltage-gated calcium channels. Thus, both SK and BK channels read calcium levels from the sub-membrane N/Q/R-type pool (`cai`) rather than the L/T-type pool (`cali`).

SK Channel (`skkca` - Small-Conductance)

- **Maximum Conductance (Somatic):** $\bar{g} = 0.145 \text{ S/cm}^2$
- **State Variables:** o (open fraction gating variable, two-state Markov scheme: Closed $\leftrightarrow$ Open)
- **Current Formula:** $i_K = \bar{g} \cdot o \cdot (V - E_K)$
- **Gating Differential Equation:**

$$\frac{do}{dt} = \frac{o_{\infty} - o}{\tau_o}$$

- **Transition Rates & Kinetic Equations:**

$$- \alpha = \frac{[\text{Ca}]_i \cdot a_{\text{bar}}}{[\text{Ca}]_i + K_1(V)}$$

$$- \beta = \frac{b_{\text{bar}}}{1 + \frac{[\text{Ca}]_i}{K_2(V)}}$$

$$- \tau_o = \frac{s_{\text{tau}}}{\alpha + \beta}$$

$$- o_{\infty} = \frac{\alpha}{\alpha + \beta}$$

- where $K_i(V) = k_i \cdot e^{-\frac{2 \cdot d_i \cdot F \cdot V}{R \cdot T}}$ are voltage-dependent dissociation constants.

Membrane Potential V (mV)	τ_m (ms)	τ_h (ms)
$V \leq -100.0$	13.33	2333.33
-95.0	15.00	2333.33
-90.0	16.27	2333.33
-85.0	18.33	2333.33
-80.0	21.47	2333.33
-75.0	25.00	2333.33
-70.0	27.97	2333.33
-65.0	30.00	2333.33
-60.0	31.17	2333.33
-55.0	31.67	2333.33
-50.0	31.80	2333.33
-45.0	32.33	2333.33
-40.0	33.07	2333.33
-35.0	31.67	2333.33
-30.0	26.57	2333.33
-25.0	20.00	2333.33
-20.0	14.83	2333.33
-15.0	11.67	2333.33
-10.0	9.77	2333.33
-5.0	8.33	2333.33
0.0	6.67	2247.50
5.0	5.00	2000.00
10.0	3.87	1580.07
15.0	3.33	1166.67
20.0	3.20	927.77
25.0	3.33	833.33
30.0	3.50	778.77
35.0	3.33	733.33
40.0	2.67	694.50
45.0	1.67	666.67
$V \geq 50.0$	1.67	666.67

- **Thermodynamic Constants:**

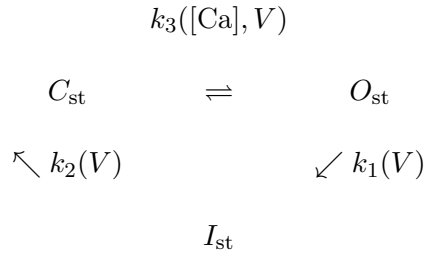
- F (Faraday constant) = 96.4853 kC/eq
- R (Gas constant) = 8.3144 J/(mol · K)
- T (Absolute temperature) = 273.15 + celsius (NEURON default 35°C → 308.15 K)

- **Channel Parameters:**

- $d_1 = 0.84$ (activation gating valence)
- $d_2 = 1.0$ (deactivation gating valence)
- $k_1 = 0.18$ mM (dissociation constant at 0 mV)
- $k_2 = 0.011$ mM (dissociation constant at 0 mV)
- $a_{\text{bar}} = 0.48 \text{ ms}^{-1}$ (maximum forward rate)
- $b_{\text{bar}} = 0.28 \text{ ms}^{-1}$ (maximum backward rate)
- $s_{\text{tau}} = 1.0$ (time constant scale factor)
- $Q_{10} = 1.0$ (temperature scaling sensitivity)

BK Channel (bkkca - Large-Conductance)

- **Maximum Conductance (Somatic):** $\bar{g} = 1.0 \times 10^{-3} \text{ S/cm}^2$
- **State Variables:** Three-state cyclic Markov kinetic model with fractions of Closed (C_{st}), Open (O_{st}), and Inactivated (I_{st}) states:



Subject to conservation: $C_{\text{st}} + O_{\text{st}} + I_{\text{st}} = 1.0$.

- **Current Formula:** $i_K = \bar{g} \cdot O_{\text{st}} \cdot (V - E_K)$
- **Transition Rate Constants:**

- k_1 (**Open** → **Inactivated**): Rapid inactivation upon depolarization.
 - * $k_1 = \text{alp}(0.1, V, -10.0, 1.0) = \frac{1}{0.1 + e^{\frac{V+10.0}{1.0}}} \cdot 1.0$
 - k_2 (**Inactivated** → **Closed**): Recovery from inactivation (favored at hyperpolarized potentials).
 - * $k_2 = \text{alp}(0.1, V, -120.0, -10.0) = \frac{1}{0.1 + e^{\frac{V+120.0}{-10.0}}} \cdot 1.0$
 - k_3 (**Closed** → **Open**): Dual voltage and cooperative calcium gating (third-order calcium dependence).
 - * $k_3 = \text{alpha}(0.001, 1.0, V, -20.0, 7.0) \cdot 1.0 \times 10^8 \cdot [\text{Ca}]_i^3$
 - * where $\text{alpha}(t_{\min}, t_{\max}, V, V_{\text{half}}, k) = \frac{1}{t_{\min} + \frac{1}{\frac{1}{t_{\max} - t_{\min}} + e^{\frac{V - V_{\text{half}}}{k}}}}$
 - k_4 (**Open** → **Closed**): Deactivation closing rate.
 - * $k_4 = \text{alp}(0.01, V, -44.0, -5.0) = \frac{1}{0.01 + e^{\frac{V+44.0}{-5.0}}} \cdot 1.0$
-

4. Submembrane Calcium Dynamics

The intracellular space contains two distinct, independent submembrane shell pools to prevent artificial crosstalk between channel classes:

1. **Standard Calcium Pool** (`cai` / `cadyn_nqr`): Coupled to high-threshold N-, P/Q-, and R-type calcium currents (i_{Ca}).
2. **L/T-type Calcium Pool** (`cali` / `cadyn_lt`): Coupled to L-type (CaV1.2, CaV1.3) and T-type calcium currents (i_{CaL}).

Each submembrane shell uses a Michaelis-Menten active extrusion pump and a passive decay process.

Dynamical Differential Equations

For the standard pool (`cai`):

$$\frac{d[\text{Ca}]_i}{dt} = \text{drive}_{\text{channel}} + \text{pump} \cdot \text{drive}_{\text{pump}} + \frac{C_{\infty} - [\text{Ca}]_i}{\tau_r}$$

For the L/T-type pool (`cali`):

$$\frac{d[\text{Ca}]_{\text{L},i}}{dt} = \text{drive}_{\text{channel}} + \text{pump} \cdot \text{drive}_{\text{pump}} + \frac{C_{\infty} - [\text{Ca}]_{\text{L},i}}{\tau_r}$$

Drive Variables & Pump Equations

- **Channel Influx:**

$$\text{drive}_{\text{channel}} = \begin{cases} \frac{-\text{drive} \cdot I_{\text{Ca}}}{2 \cdot F \cdot \text{depth} \cdot F_{\text{spine}}} & \text{if } I_{\text{Ca}} < 0 \\ 0 & \text{if } I_{\text{Ca}} \geq 0 \end{cases}$$

- **Michaelis-Menten Extrusion Pump:**

$$\text{drive}_{\text{pump}} = \frac{-k_t \cdot [\text{Ca}]_i}{[\text{Ca}]_i + k_d}$$

- **Dynamics Parameters:**

- C_{∞} (equilibrium baseline concentration) = 1.0×10^{-5} mM (10 nM)
 - τ_r (exponential decay removal time constant) = 43 ms
 - drive (influx scaling coefficient) = 10000
 - F (Faraday constant) = 96489 C/mol
 - depth (submembrane shell depth) = 0.1 μm
 - F_{spine} (spine factor) = 1.0
 - k_t (maximum pump transport rate) = 1.0×10^{-4} mM/ms
 - k_d (pump dissociation constant) = 1.0×10^{-4} mM
 - pump (extrusion multiplier scaling factor) = 0.02
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5. Synaptic Receptors

Excitatory (glutamatergic AMPA and NMDA) and inhibitory (GABAergic) inputs are modeled using dual-exponential conductances.

AMPA Synapses (Excitatory)

- **Model Type:** Dual-exponential conductance ($g = h - m$)
- **Rise Time Constant (τ_1):** 1.1 ms
- **Decay Time Constant (τ_2):** 5.75 ms

- **Reversal Potential (e):** 0 mV

- **State Equations:**

$$\begin{aligned}
- \quad m' &= -\frac{m}{\tau_1}, \quad h' = -\frac{h}{\tau_2} \\
- \quad \text{NetReceive: } m &= m + w \cdot f_{\text{factor}}, \quad h = h + w \cdot f_{\text{factor}} \\
- \quad \text{where } t_p &= \frac{\tau_1 \cdot \tau_2}{\tau_2 - \tau_1} \cdot \ln\left(\frac{\tau_2}{\tau_1}\right) \text{ and } f_{\text{factor}} = \frac{-1}{e^{-t_p/\tau_1} - e^{-t_p/\tau_2}}
\end{aligned}$$

NMDA Synapses (Excitatory with Magnesium Block)

- **Model Type:** Dual-exponential conductance with voltage-dependent magnesium blocking factor $B(V)$ ($g = (h - m) \cdot B(V)$)
- **Rise Time Constant (τ_1):** 2.82 ms
- **Decay Time Constant (τ_2):** 160 ms
- **Reversal Potential (e):** 0 mV
- **External Magnesium Concentration ($[\text{Mg}^{2+}]_o$):** 1.0 mM
- **Magnesium Block Function:**

$$B(V) = \frac{1}{1 + \frac{[\text{Mg}^{2+}]_o}{3.57} \cdot e^{-V \cdot 0.062}}$$

GABA Synapses (Inhibitory)

- **Model Type:** Dual-exponential conductance ($g = h - m$)
- **Rise Time Constant (τ_1):** 0.25 ms
- **Decay Time Constant (τ_2):** 3.75 ms
- **Reversal Potential (e):** -60 mV

6. Morphology & Compartmental Properties

To incorporate dendritic spines without explicitly modeling them as separate compartments, the passive membrane area (capacitance C_m and leak conductance g_{pas}) is scaled up by a **Spine Factor** (F_{spine}). To preserve the original input resistance and axial electrotonic cable properties, dendritic dimensions are scaled by:

- $d_{\text{adjusted}} = d_{\text{true}} \cdot F_{\text{spine}}^{3/2}$
- $L_{\text{adjusted}} = L_{\text{true}} \cdot F_{\text{spine}}^{1/2}$

Morphological Cable Dimensions

Section	Branches	Segments	l (μm)	d (μm)	F_{spine}	l' (μm)	d' (μm)
Soma	1	1	16.0	16.0	1.00	16.00	16.00
Primary Dendrites	4	1	20.0	2.25	1.00	20.00	2.25
Secondary Dendrites	8	1	20.0	1.00	1.33	24.23	1.10
Tertiary Dendrites	16	11	190.0	0.50	3.00	395.20	0.72

Specific Membrane Properties

- **Axial Resistivity** (R_a): $100 \Omega \cdot \text{cm}$
- **Specific Capacitance** (C_m): $1.0 \mu\text{F}/\text{cm}^2$ (scaled dynamically by F_{spine} in dendrites)
- **Passive Leak Reversal** (E_{pas}): -70.0 mV
- **Passive Leak Conductance** (g_{pas}): $11.5 \times 10^{-6} \text{ S}/\text{cm}^2$ (scaled dynamically by F_{spine} in dendrites)

Compartmental Conductance Densities

Channel conductances are distributed differentially across the compartmental tree:

- Active conductances ($\bar{g}$) are in units of **S/cm²**.

- Calcium permeabilities ($\bar{p}$) are in units of **cm/s**.

Synapse Spatial Distribution

Colocalized AMPA/NMDA (excitatory glutamatergic) synapses are targeted at spines, while GABA (inhibitory GABAergic) synapses are targeted at the somatic and dendritic shafts.

- **Glutamatergic Synapses (AMPA/NMDA):** 84 total locations
 - **Primary Dendrites:** 1 location per branch at 0.5 (4 locations total)
 - **Secondary Dendrites:** 2 locations per branch at 0.33, 0.66 (16 locations total)
 - **Tertiary Dendrites:** 4 locations per branch at 0.2, 0.4, 0.6, 0.8 (64 locations total)
 - **GABAergic Synapses (GABA):** 80 total locations
 - **Soma:** 16 locations at 0.5 (16 locations total)
 - **Primary Dendrites:** 3 locations per branch at 0.25, 0.5, 0.75 (12 locations total)
 - **Secondary Dendrites:** 3 locations per branch at 0.25, 0.5, 0.75 (24 locations total)
 - **Tertiary Dendrites:** 2 locations per branch at 0.33, 0.66 (32 locations total)
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7. Experimental & Simulation Setups

Standard Simulation Parameters

- **Initial Voltage (V_{init}):** -87.75 mV (reflecting hyperpolarized biological "down-state" resting potential)
- **Temperature:** 35°C (`h.celsius`)
- **Standard Time Step (dt):** 0.01 ms

Mechanism	Soma	Primary	Secondary	Tertiary
Leak (g_{pas})	11.5×10^{-6}	11.5×10^{-6}	11.5×10^{-6}	11.5×10^{-6}
NaF ($\bar{g}_{\text{naf}}$)	1.50	0.0195	0.0195	0.0195
NaP ($\bar{g}_{\text{nap}}$)	4.0×10^{-5}	1.3802×10^{-7}	1.3802×10^{-7}	1.3802×10^{-7}
KAf ($\bar{g}_{\text{kaf}}$)	0.225	0.225	0.020584	0.020584
KAs ($\bar{g}_{\text{kas}}$)	0.0104	0.0104	9.5142×10^{-4}	9.5142×10^{-4}
Kir ($\bar{g}_{\text{kir}}$)	1.4×10^{-4}	1.4×10^{-4}	1.4×10^{-4}	1.4×10^{-4}
Krp ($\bar{g}_{\text{krp}}$)	0.001	0.0	0.0	0.0
CaL12 ($\bar{p}_{\text{cal12}}$)	6.7×10^{-6}	6.7×10^{-6}	6.7×10^{-6}	6.7×10^{-6}
CaL13 ($\bar{p}_{\text{cal13}}$)	4.25×10^{-7}	4.25×10^{-7}	4.25×10^{-7}	4.25×10^{-7}
CaN ($\bar{p}_{\text{can}}$)	1.0×10^{-5}	1.0×10^{-5}	1.0×10^{-5}	1.0×10^{-5}
CaR ($\bar{p}_{\text{car}}$)	2.6×10^{-5}	2.6×10^{-5}	2.6×10^{-5}	2.6×10^{-5}
CaQ ($\bar{p}_{\text{caq}}$)	6.0×10^{-6}	6.0×10^{-6}	6.0×10^{-6}	6.0×10^{-6}
CaT ($\bar{p}_{\text{cat}}$)	4.0×10^{-7}	4.0×10^{-7}	4.0×10^{-7}	4.0×10^{-7}
BK ($\bar{g}_{\text{bkkea}}$)	0.001	0.001	0.001	0.001
SK ($\bar{g}_{\text{skkea}}$)	0.145	0.145	0.145	0.145

- **Standard Resolution:** 100 points/ms (`steps_per_ms`)
- **Ion Initial & Boundary Concentrations:**
 - $[\text{Ca}]_i^0$ (initial internal standard calcium) = 0.001 mM (1 μM)
 - $[\text{Ca}]_o^0$ (external standard calcium) = 5.0 mM
 - $[\text{Ca}]_{\text{L},i}^0$ (initial internal L/T-type calcium) = 0.001 mM (1 μM)
 - $[\text{Ca}]_{\text{L},o}^0$ (external L/T-type calcium) = 5.0 mM

Current Clamp Protocols

- **Somatic Injection Sweeps (`IClamp.py`):**
 - Stimulus duration = 500.0 ms
 - Stimulus onset delay = 200.0 ms
 - Total simulation time = 900.0 ms
 - Injection amplitudes tested:
 - * +0.271 nA: Triggers action potential spike trains.
 - * +0.248 nA: Triggers near-threshold subthreshold depolarizations.
 - * -0.227 nA: Triggers hyperpolarization to characterize strong inward rectification (Kir).
- **Delayed Excitation & KAs Modulation Sweep (`chan_currents.py`):**
 - Stimulus amplitude = +0.232 nA
 - Stimulus duration = 600.0 ms
 - Stimulus onset delay = 100.0 ms
 - Total simulation time = 800.0 ms
 - Condition sweeps scale the slow KAs inactivation variable `a_kas` by factors of 1.0 (Control), 0.6 (decreased delay), and 1.4 (increased delay) to study its role in late-firing action potentials.
- **I-V Relationship Sweep (`current_volt_freq.py`):**
 - Stimulus amplitude sweeps: 20 linear increments between -0.2 nA (-200 pA) and +0.2 nA (+200 pA).
 - Stimulus duration = 500.0 ms
 - Stimulus onset delay = 100.0 ms

- Somatic potential measured at $t = 550.0$ ms (450 ms after stimulus onset, to capture steady-state rectifying behavior).
- **F-I Firing Frequency Sweep** (`current_volt_freq.py`):
 - Stimulus amplitude sweeps: 15 linear increments between +0.15 nA and +0.45 nA.
 - Stimulus duration = 500.0 ms
 - Stimulus onset delay = 100.0 ms
 - Spike frequency calculated as $F = \frac{N_{\text{spikes}}}{0.5 \text{ s}}$ for spikes occurring during the injection window (100.0 ms to 600.0 ms).